EE3032 Introduction to VLSI Design—Final

1. (20 points) Figure 1 shows an NP domino logic circuit. Assume that $\mu_n = 2\mu_p$, $CLK \times \overline{CLK} = 0$, and the rise time and fall time of the clock signal can be neglected. Also, a $2\mu m/0.18\mu m$ PMOS/NMOS has an input gate capacitance C; the turned on resistance of a $2\mu m/0.18\mu m$ NMOS is R; and the parasitic capacitance of all the transistors can be neglected. Complete the following questions.

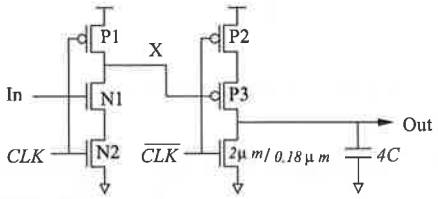


Figure 1: An NP domino logic circuit

   (a) Write out the Boolean function of the NP domino logic circuit and explain the operation principle of the circuit. (5 points)

   (b) Assume that the duty cycle of CLK is 50%, $t_r = t_f$ for the nodes X and Out, $(W/L)_{N1} = (W/L)_{N2}$, and $(W/L)_{P2} = (W/L)_{P3}$. Determine the minimum clock period of CLK in terms of R and C (4 points); and the sizes of N1, P1, and P2 (6 points).

   (c) Assume the supply voltage is V_{dd}. Calculate the switching power consumed by the clocks CLK and $\overline{CLK}$. (5 points)

2. (8 points) Figure 2 shows a Schmitt trigger circuit, where $V_{tn} = |V_{tp}| = 0.6V$. If you apply an input signal as shown in the left figure to the Schmitt trigger circuit, then the circuit outputs the signal as shown in the right figure. Complete the following questions.

   (a) Derive the ratios of the $\frac{\beta_{N_1}}{\beta_{N_3}}$ and $\frac{\beta_{P_1}}{\beta_{P_3}}$. (4 points)

   (b) In comparison with a typical inverter (i.e., without the transistors P_3 and N_3), is the short-circuit power of the Schmitt trigger circuit higher or smaller than that of the typical inverter? Why? (4 points)

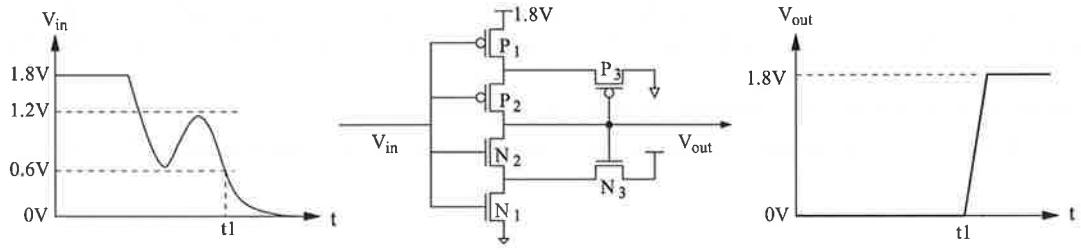


Figure 2: A CMOS Schmitt trigger circuit

3. (10 points) Figure 3 shows a domino logic gate. Assume that $0.18\mu m$ technology is used to design the domino logic gate and $\mu_n = 2\mu_p$. Also, a $2\mu m/0.18\mu m$ PMOS/NMOS has an input gate capacitance C. Complete the following questions.

   (a) Assume that the channel length of all the transistors is $0.18\mu m$. If we desire to design the domino logic gate such that the switching power of the clock (CLK) is $9fCV_{dd}^2$, where f and V_{dd} denote the operation frequency and the power supply of the logic gate. Determine the sizes of N1, N2, N3, N4, N5, N6, N7, and P1 such that the rise time and the fall time of the node X are equal, i.e., $t_r = t_f$. (5 points)

   (b) The domino logic may have a malfunction due to the leakage current when the state of X is logic 1. Modify the domino logic gate shown in Figure 3 such that it can be functional correct. (5 points)

4. (15 points) Draw the transistor-level schematic diagrams of the two symbolic layouts shown in Figure 4.

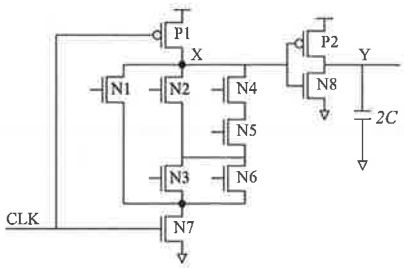


Figure 3: A domino logic gate

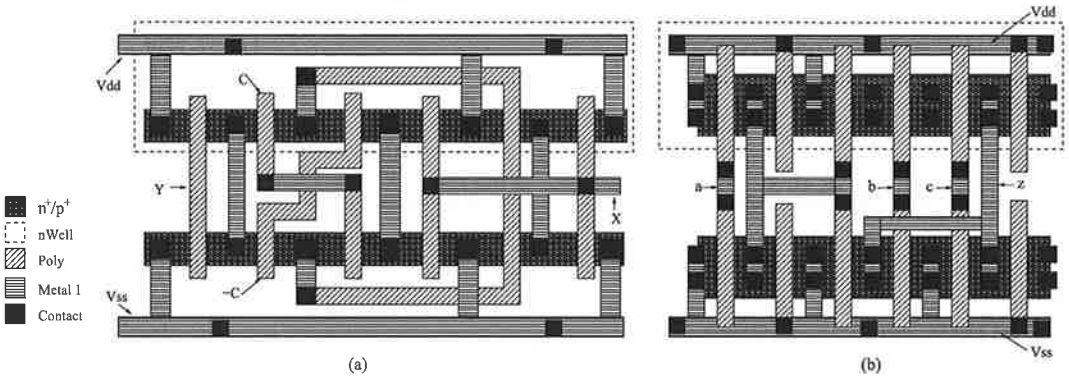


Figure 4: Symbolic layouts of two circuits

5. (15 points) Figure 5 shows a sequential circuit. Assume that the clock-to-Q propagation delay (t_{pcq}), the clock-to-Q contamination delay (t_{ccq}), the setup time (t_{setup}), and the hold time (t_{hold}) of the registers are $0.3ns$, $0.1ns$, $0.3ns$, and $0.4ns$, respectively. The delay of the NOR gate, the XOR gate, the XNOR gate, and the inverter is $0.3ns$, $0.5ns$, $0.4ns$, and $0.2ns$, respectively. Also, the delay of interconnection between R1 and R4 is $0.1ns$. Complete the following questions.

   (a) Can the circuit be operated correctly? If not, how do you modify the circuit by inserting additional logic gates (e.g., the NOR gate, XONOR gate, or the inverter shown above) such that it can operate correctly without sacrifice the operation frequency? Draw the modified circuit. (6 points)

   (b) Determine the maximum operation frequency of the circuit. (4 points)

   (c) If the clock skew is $0.5ns$, determine the maximum operation frequency of the sequential circuit. (5 points)

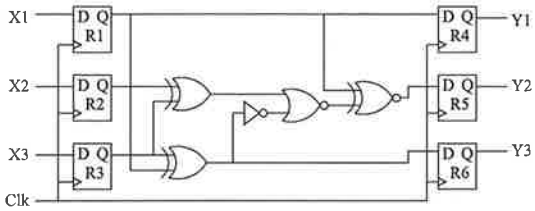


Figure 5: A sequential circuit

6. (10 points) Figure 6 shows one possible implementation of the D register. Complete the following questions.

   (a) Figure 6 shows a register consisting of a master latch and a low-cost slave latch. Assume that the delay of Inv1, Inv2, and transmission gate is 0.2ns, 0.2ns, and 0.1ns, respectively. Define the value of setup time of the register if the clock is ideal. (4 points)

   (b) As Figure 6 shows, assume that the switching threshold voltage of the inverter Inv2 is $V_{dd}/2$ and the turn-on equivalent resistances of the transmission gate, N1, N2, P1, and P2 are R_{TS}, R_{N1}, R_{N2}, R_{P1}, and R_{P2}, respectively. List two inequalities for determining the sizes of P2 and N2 such that the register can be operated correctly. (6 points)

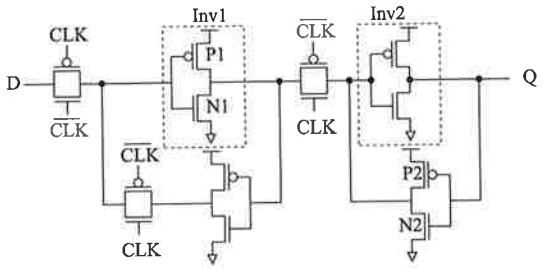


Figure 6: A static register

7. (12 points) Sketch the Boolean function $f = \overline{(A + BC)D}$ using each of the following logic design techniques: (1) Dynamic logic (4 points) (2) Fully differential logic (4 points) (3) Differential split-level logic (4 points).

8. (30 points) Answer the following questions.

   (a) List two approaches which can be used to alleviate the occurrence probability of latchup. (4 points)

   (b) List two requirements which must be met when you draw the layout of a cell in the standard cell library. (4 points)

   (c) Why is the soft error rate of a circuit increased with the increasing of operation frequency of the circuit and the scaling down of technology process? Explain the cause in terms of the following three effects: logical masking effect, electrical masking effect, and latch window effect. (6 points)

   (d) List two methods to reduce the power bounce and ground bounce occurring in the IO pads. (4 points)

   (e) List two major problems of the dynamic logic. (4 points)

   (f) List two limitations of the CMOS domino logic. (4 points)

   (g) What are the core-limited design and the pad-limited design? (4 points)